

Anubhav Joshi

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SUMMARY

Backend and AI engineer building scalable backend systems with very deep and complex AI pipelines, both in my recent startup experience and in my personal projects. I have worked with OCR and OpenCV, built a custom RAG system specifically to power my LLMs with hand-curated knowledge base grounding, and built an intermediary security and governance SDK for AI agents and the tools they are allowed to use.

SKILLS

Languages: Python, TypeScript, Rust

AI/ML: PyTorch, LangChain, LangGraph, Google ADK

Backend and System Design: FastAPI, PostgreSQL, pgvector, Pinecone, CockroachDB, Redis, Docker, AWS, Google Cloud, Vercel, Render

UI/UX: Next.js, React Native

EXPERIENCE

Co-Founder and CTO, Anchorate - Jaipur, India

Jan 2026 - Aug 2026

- Built Anchor8, a governance layer that inspects every tool call an autonomous AI agent makes, catching leaked PII, prompt injections, and destructive actions before they execute. Published on PyPI as **`pip install anchor8`**.
- Designed its 3-lane pipeline: deterministic checks first (PII regex, injection patterns, Z-score anomaly detection in Redis, pgvector similarity against the agent's own history), then a two-LLM adjudication step for the ambiguous middle.
- Shipped SDKs in Python, TypeScript, and Go with a 3-line LangChain integration, leading a team of 7. The experience made me grow as an engineer.

SDE Intern, Blinkadz - Jaipur, India

Feb 2025 - Apr 2025

- Built an end-to-end video ad generation pipeline on Google ADK weeks after its release, working out the orchestration patterns by experiment rather than from docs, and integrated LinkedIn's marketing APIs for ads and campaigns.

PROJECTS

Spoin: For the Curious, Python, FastAPI, LangGraph, PostgreSQL, pgvector

Sole architect, 103 ADRs

- Made a probabilistic system behave deterministically by grounding everything it would otherwise guess: text retrieves from a hand-curated knowledge base, photographs are assigned from an open-licensed catalog by retrieval instead of generated, and ASCII diagrams are verified server-side against a fixed geometry contract.
- Wrote a load balancer that schedules every generation call across a per-key-per-model quota grid, reaching **499+ items/min peak throughput on nothing but free-tier Gemini keys**.
- Built frog, a custom RAG package written for this pipeline, owning the embedder (Qwen3-Embedding-0.6B at 1024 dimensions, GPU for bulk work, CPU for the request path), chunking, and retrieval. It runs in-process, so planner, drafter, and verifier each retrieve per call with no HTTP hop.
- Every card records the corpus chunks that produced it, so any claim is traceable to its source. Grounded runs deliver 523 cards in 21 minutes at 24.85 cards/min with zero HTTP 429s, down from 917 on the previous run, which hit IP-side rate limiting.
- Designed the knowledge base, the_spoin_universe, as a claim-qualified knowledge system where every claim carries type, version, and provenance, building toward a semantic graph over everything it accumulates.

Fraud Vote Detector, Python, OpenCV, Google Cloud Vision OCR

github.com/anubhav-qt/fraud-vote-detection

- Audits scanned PDF electoral rolls deterministically: OpenCV segments each voter card, OCR reads the fields at 98%+ accuracy across Hindi and English, and face encodings flag the same person registered twice above 90% similarity.

Synthetic Medical Dataset Generator, PyTorch, DCGAN

github.com/anubhav-qt/synthetic-dataset-generator

- A 7-layer DCGAN trained 125 epochs on NIH chest X-rays, generating 256x256 synthetic scans for rare pulmonary conditions, with label smoothing, noise injection, and gradient clipping holding off mode collapse at FID 150 to 180.

EDUCATION

Manipal University Jaipur, Jaipur, India

Aug 2022 - Jul 2026

B.Tech in Computer Science, specialization in Artificial Intelligence and Machine Learning | CGPA: 9.28/10

Dean's List and Student Excellence Award, all 8 semesters. Vice Chair, IEEE GRSS student chapter.